MIL-C-82075B(YD)
28 May 1981
SUPERSEDING
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14 May 1976

MILITARY SPECIFICATION

CAMEL AND SEPARATOR, MARINE (FOR SUBMARINE BERTHING)

This specification is approved for use by the Naval Facilities Engineering Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two styles of prefabricated, knocked-down, timber berthing structures.

1.2 Classification. The structure shall be one of the following styles, as specified (see 6.2).

Style A - Camel.
Style B - Separator.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-781 - Strapping, Steel, and Seals.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner.

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-C-17415 - Cloth, Coated, and Webbing, Inflatable Boat and Miscellaneous Use.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 1945

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STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

BUREAU OF YARDS AND DOCKS (NAVAL FACILITIES ENGINEERING COMMAND)

1028247 - Camels and Separators for Nuclear Submarines Design, Structural.
1028248 - Camels and Separators for Nuclear Submarines Details, Structural.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN INSTITUTE OF STEEL CONSTRUCTION (AISC)

Specification for the Design Fabrication and Erection of Structural Steel for Buildings.

(Application for copies should be addressed to the American Institute of Steel Construction, Inc., 101 Part Avenue, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A36 - Specification for Structural Steel.
A153 - Specification for Zinc, Coating (Hot-Dip) on Iron and Steel Hardware.
B127 - Specification for Nickel-Copper Alloy Plate, Sheet, and Strip.
B164 - Specification for Nickel-Copper Alloy Rod and Bar.
D1785 - Poly(Vinyl Chloride) (PVC) Plastic Pipe, Schedules 40, 80, and 120.
D2000 - Rubber Products in Automotive Applications, Standard Classification System.
D2240 - Rubber Property - Durometer Hardness, Test for.
D2464 - Threaded Poly(Vinyl Chloride) (PVC) Plastic Pipe Fittings, Schedule 80.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

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AMERICAN WOOD-PRESERVERS' ASSOCIATION (AWPA)

A2 - Analyses of Water-Borne Preservatives and Fire-Retardant Formulations.
A6 - Determination of Water and Oil-Type Preservatives in Wood.
C1 - All Timber Products - Preservative Treatment by Pressure Processes.
C18 - Standard for Pressure Treated Material in Marine Construction.
M2 - Inspection of Treated Timber Products.

(Application for copies should be addressed to the American Wood Preservers Association, 7735 Old Georgetown Rd., Bethesda, MD 20014.)

Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Description. Camel and separator berthing structures shall be furnished knocked-down and shall include all prefabricated members, parts, fasteners, and materials shown and specified on Drawings 1028247 and 1028248, and as specified herein. The structures shall be timber frame, triangular in shape, rigidly constructed, when assembled, to withstand stresses without damage or permanent deformation due to berthing operations of submarines. All camels and separators shall be completely prefabricated, assembled, inspected, and then knocked-down and prepared for shipment. Each structure shall be furnished with a 1,000 gallon rubber or plastic buoyancy tank assembly. The tank material and construction shall be capable of continuous use in water and air, at temperatures from -40° Fahrenheit (F) to 210°F, and shall be resistant to sea water, all fuel oils, and abrasion. Concrete ballast will be furnished and installed by others during final assembly on site.

3.1.1 Style A. The camel shall include timber fenders on the top and bottom of one side and wing type rubber fenders on the corner post opposite in accordance with Drawing 1028248.

3.1.2 Style B. The separator shall include wing type rubber fenders on corner posts in accordance with Drawing 1028248.

3.2 First article. When specified (see 6.2), the contractor shall furnish a camel or separator for first article inspection and approval (see 4.2.1 and 6.3).

3.3 Identical items. All berthing structures of the same style furnished under a specific contract shall be physically and mechanically identical, in that all like prefabricated members, parts, assemblies, and fasteners shall be interchangeable with those of all other berthing structures produced. Written approval for deviations must be obtained in advance from the contracting officer.

3.4 Drawings. The drawings, forming a part of this specification, are engineering design drawings. The contractor is responsible for preparing his own shop drawings. No deviations from the prescribed dimensions is

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permissible without prior written approval of the contracting officer. In the event of a conflict between requirements on drawings and requirements specified herein, the specification shall have precedence over the drawings.

* 3.5 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

* 3.5.1 Timber. Timber shall be coastal Douglas Fir conforming to the structural requirements stated on NAVFAC drawing no. 1028247. With the contracting officer's prior approval, the contractor may use southern pine as substitute timber, provided it meets the structural requirements of drawing no. 1028247. All timbers shall be treated in accordance with the requirements of AWPA C1 and C18 for structural lumber in salt water. Treatment shall be as specified (see 6.2 and 6.4). All timbers intended for the same purpose shall be of the same species.

3.5.2 Steel. Structural steel shapes and plate shall conform to ASTM A36.

3.5.3 Monel alloy. Monel alloy shall conform to ASTM B127 and B164 as applicable, and shall have a hardness of not less than 90 on the Rockwell B scale.

* 3.5.4 Rubber, shaped. Rubber shaped by extrusion shall be styrene-butadiene in conformance to ASTM D2000, material designation 4AA720A13, or equivalent synthetic rubber of similar physical properties.

3.5.5 Tank fabric. At the option of the contractor, the tank fabric shall be either rubber impregnated cotton or nylon cloth, or laminated plastic material. Cloth and coatings, type and class of nylon and cotton coated fabrics, shall conform to MIL-C-17415, and shall be as specified in table I, II, and III therein. The fabric shall have strength, weight, and coatings sufficient to meet requirements for the completed fabric tank specified herein and on the drawings and tests in 4.4. Plastic material shall be a laminated plastic having characteristics suitable for the purpose intended.

3.5.6 Plastic pipe. Pipe shall be poly(vinyl chloride) (PVC) plastic in conformance to ASTM D1785. Pipe shall be schedule 80, threaded design.

3.5.7 Valves and fittings. Valves and fittings shall be PVC plastic in conformance to ASTM D2464.

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3.6 Construction.

3.6.1 Connecting plates and lifting lugs. Fabrication of connecting plates and lifting lugs shall be in accordance with AISC Specification for the Design, Fabrication, and Erection of Structural Steel for Buildings. The forming of material shall be done by methods that will not cause injury to the metal.

3.6.2 Rubber fenders. Rubber fenders shall be the wing type, fabricated as shown on the drawings. The hardness of the fender shall be 70 ±5 of the Shore A durometer (see 4.5.1.2). The tensile strength shall be at least 2,000 pounds per square inch (psi). All shaping and fabrication of the extrusion shall be done prior to the standard curing process. The fender shall be resistant to sea water, fuel oils, weathering, and abrasion, and shall be flexible between temperatures of -40°F and 210°F. Standard commercial products may be used provided they meet the requirements specified herein.

3.6.3 Fasteners and hardware. Bolts, nuts, washers, eye bolts, and lag screws shall be zinc-coated, standard commercial type, conforming to ASTM A153, class C. Metal flange fabric reinforcements, pipe guide, and capscrews shall be monel metal or equivalent corrosion-resistant metal.

3.7 Workmanship.

3.7.1 Timber. All timber shall be prefabricated for assembly, complete with bolt holes, before the pressure treatment.

3.7.2 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing process shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.7.3 Bolted connections. Bolt holes shall be accurately punched or drilled to coincide with bolts of the connection in accordance with requirements of AISC Specification for the Design, Fabrication and Erection of Structural Steel for Buildings, and shall have the burrs removed.

3.7.4 Welding. Welding procedure shall be in accordance with a nationally recognized welding code. Surfaces to be welded shall be free from rust, loose scale, paint, grease, or other foreign matter. Welds shall transmit maximum imposed stresses without permanent deformation or failure.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein, and complete inspection records shall be furnished to the contracting officer. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections

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are deemed necessary to assure supplies and services conform to prescribed requirements. Tests to verify the accuracy of inspection reports furnished by the contractor shall be made either by the contracting officer or by commercial inspection companies retained by the contracting officer. The contracting officer may elect to employ the services and accept the stamp of an independent quality control agency.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).
(c) Packaging inspection (see 4.5).

4.2.1 First article. First article inspection shall be performed on one camel or separator without the tank assembly installed, when a first article sample is required (see 3.2 and 6.2). The inspection shall include the examination of 4.3 and the tests of 4.4. Approval of the first article by the activity concerned shall not relieve the contractor of his obligation to supply camels and separators conforming to this specification. Any changes or deviations of the camels and separators from the first article shall be subject to the approval of the contracting officer.

4.2.2 Quality conformance inspection. Quality conformance inspection shall be performed on each unit in accordance with examinations of 4.3 through 4.3.2.

4.3 Examination. Each camel and separator shall be examined for compliance with the requirements specified in Section 3 of this specification. The element of inspection shall encompass all visual examinations and dimensional measurements. Noncompliance with any specified requirement or presence of one or more defects preventing or lessening maximum efficiency shall constitute cause for rejection. In addition, all tanks shall be tested as specified in 4.4.2.

4.3.1 Verification of preservative treatment. Treated wood shall be inspected in accordance to AWPA M2 with the following exceptions. Twenty pieces of a lot shall be bored for measurement of penetration. Southern pine shall be bored from sapwood faces only. Douglas fir shall be bored from heartwood faces only. If 80 percent or more of the borings meet the penetration requirements, the lot is considered as meeting penetration requirements. For the assay of treated southern pine lumber or timber, a composite sample of the outer 0.60 inch of borings shall be taken from sapwood. For the assay of Douglas fir lumber or timber, a composite sample of the outer 0.60 inch of borings shall be taken from heartwood. Single borings shall be taken from 48 representative pieces treated with creosote or creosote-containing solutions, and from 20 representative pieces treated with waterborne preservatives.

4.3.2 Assay. The determination of the amount of creosote or creosote-containing solutions in a sample of borings shall be made according to AWPA A6. The determination of the amounts of waterborne preservatives in a sample of borings shall be made according to AWPA A2. Retention of 90 percent of the stipulated assay retention will be accepted as conforming to stated requirements.

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4.4 Tests.

4.4.1 First article tests.

4.4.1.1 Test erection unit. Parts and components of one structure, excluding the tank assembly, shall be assembled for inspection in accordance with Section 4.3 to determine compliance with Section 3. Any major defect shall cause all camels and separators in the contract to be tested as specified herein. The contractor shall provide all labor, facilities, and supplies necessary for test erection assembly and disassembly, including any repair or replacement parts necessary. The test erection structure shall then be packed as part of a completed structure.

4.4.1.2 Hardness test. Test for hardness of rubber fenders shall be made in accordance with ASTM Designation D2240. Hardness value other than that specified in 3.6.2 shall be cause for rejection.

4.4.2 Tank tests. All tanks shall be tested while supported vertically in a rack which affords support in a manner similar to that of the actual berthing structure.

4.4.2.1 Hydrostatic test. All tanks shall be subjected to a hydrostatic pressure of 75 psi and shall withstand this pressure for a period of 1 hour. Evidence of leakage shall constitute failure of this test.

4.4.2.2 Air test. All tanks shall be internally pressurized at 10 psi and allowed to stand for 1 hour. All fittings, flanges, and seams shall be covered with soapsuds solution and the tank examined for leaks. Any evidence of leakage shall constitute failure of this test.

4.4.2.3 Performance. All tanks when tested by water and air shall maintain the cylindrical shape and approximate diameter and length of the design fabrication and there shall be no appreciable longitudinal or diagonal expansion, nor permanent distortions of the tank body.

4.5 Packaging inspection. The preservation-packaging, packing, and marking of the camels and separators shall be inspected to verify conformance to the requirements of Section 5.

5. PACKAGING

5.1 Preservation. The preservation shall be level A or commercial, as specified (see 6.2).

5.1.1 Level A. Unpainted and uncoated ferrous metal surfaces, and the threaded surfaces of fasteners shall be coated with type P-1 preservative. The preservative shall conform to the applicable specification listed in, and shall be applied in accordance with MIL-P-116.

5.1.2 Commercial. The complete camels and separators shall be preserved and packaged in accordance with the contractor's standard practice in a manner to prevent deterioration and damage during shipment.

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5.2 Packing. The packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Tanks. The tanks shall be collapsed and attached fittings and hardware shall be covered or protected with cushioning material secured in place with tape. The tanks shall be packed in close-fitting boxes conforming to PPP-B-601, military overseas type; or PPP-B-621, class 2.

5.2.1.2 Connection plates and fasteners. The connection plates, lifting lugs, bolts, nuts, and washers shall be packed in containers as specified for the tanks. The contents shall be cushioned, blocked, and braced to prevent movement inside the container and damaging of the contents. When packed together, bolts shall be separated from other items in the container with a wood separator.

5.2.1.3 Timber and fenders. Timber shall be bundled in secured lifts of a weight and size convenient for handling and stowing. The bundles shall be secured with 0.035 inch by 1-1/4 inch flat steel strapping conforming to QQ-S-781, class 1, type I, finish B. The rubber fenders shall be bundled as specified for timber, except that the strapping shall be 3/4 inch by 0.023 inch.

5.2.2 Level B. Each camel and separator shall be packed as specified for level A, except that the boxes shall be normal overseas type or class 1, as applicable and strapping shall be either finish A or B.

* 5.2.3 Commercial. The camels and separators shall be packed in a manner which will insure arrival at destination in satisfactory condition. Packing shall conform to applicable carrier rules and regulations.

5.3 Marking. The shipping containers and bundles shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Camel and separator berthing structures are to be used in berthing submarines.

* 6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Style of berthing structure required (see 1.2).
(c) When first article of camels and separators is required for inspection and approval (see 3.2, 4.2.1, and 6.3).
(d) Type of preservative treatment required (see 3.5.1 and 6.4).
(e) Level of preservation and level of packing required (see 5.1 and 5.2).

* 6.2.1 Contract data requirements. When this specification is used in an acquisition which incorporates a DD Form 1423 and invokes the provision of paragraph 7-104.9(n) of the Defense Acquisition Regulations (DAR), the data

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Description (DD Form 1664) and delivered in accordance with the Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR 7-104.9(n) are not invoked, the data shall be delivered in accordance with the contract requirements.

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of paragraph 7-104.55 of the DAR. The first article should be a first production item. The contracting officer should include specific instructions in all acquisition instruments, regarding arrangements for examinations, test, and approval of the first article.

* 6.4 Treatment selection. Treatment selection, as described in AWPA C18, is based on the type of marine borer activity expected at the location of the structure's installation. The procuring office should make a determination of the treatment requirements based on the expected marine borer hazard and provide this information under 6.2.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-82075B(YD) CAMEL AND SEPARATOR, MARINE (FOR SUBMARINE BERTHING)

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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